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Lost or Wrong?

A causal runtime router that distinguishes search failure from wrong-fix failure in coding agents

失之搜索，还是错于修正？
一个判断编码智能体如何失败的运行时路由器

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A causal runtime router that distinguishes search failure from wrong-fix failure in coding agents

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Abstract

When a coding agent stops making progress, two interventions are available: help it find the code, or make it re-examine its own fix. They are opposite, and today’s runtime monitors cannot choose between them — they detect *that* an agent is struggling and carry no signal about *how*. On identical rows and folds the published stagnation- and loop-detection families score **0.554–0.607** at predicting failure, and at the question that decides the intervention — is the agent LOST (has not found the code) or WRONG-FIX (found it, and the change does not work) — they score **at or below chance (0.41–0.54)**.

We present **Route**, a causal, reference-free runtime router that reads only the first 10–60% of a run and outputs both whether the run will fail and which failure mode it is in, with the intervention that follows. Trained on one set of runs and tested on **two disjoint sets it never saw** (24 cells: every ordered pair of shard sets $\times$ four prefix fractions) it reaches **0.7145** at failure prediction and **0.7324** at mode discrimination — **+0.156 and +0.136 AUC over the published detector family** on the same rows and folds. Cross-set performance equals within-set performance, so the margin is not fitted to one corpus.

Two further results bound the claim. The **false-alarm rate is controlled** at every level ($\alpha = 0.05 \rightarrow 0.046$ achieved, 8 of 8 levels) against an explicit null, replacing an earlier claim of ours that was **circular** and is withdrawn. As a decision rule the router beats every fixed policy: **0.764–0.800** against 0.732 for always-verify, 0.656 for random and 0.518 for always-search, in **12 of 12** cells.

A live evaluation exposes a benchmark artefact: on our suite **every run reached the correct file**, because the verifier prints the failing test’s filename, which contains the module name — localisation was solved by string-matching, not diagnosis. Withholding which tests failed cuts success by **27.5 points** ($p = 0.015$) and is the only significant effect in that experiment. We also report where the method **stops** working: a model trained on SWE-agent scores 0.32–0.50 on 88,000 runs from three other scaffolds, against 0.65–0.76 for the same features fitted inside those scaffolds, so the generality established here is *shard-level within a scaffold*.

Keywords: coding agents, runtime monitoring, failure modes, intervention, reproducible research.

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1 The intervention problem: LOST and WRONG-FIX

A coding agent has run for fifty steps without solving the task, and a decision is required. The literature offers many ways to notice this — loop detectors, redundancy measures, stagnation triggers, output-shape anomalies — and they are decent at noticing: published detectors sit around AUC 0.6–0.7 [1], and a failure-as-a-process monitor reaches 82% precision at a 2–3% false-alarm rate [2].

Detecting stagnation is not deciding what to do. A stalled agent has two distinct failure modes:

- **LOST** — it has not found the code that needs changing. Long searching, wrong-file edits, or no edit at all. The remedy is to supply a location.
- **WRONG-FIX** — it found the right code and its change does not work. Repeated edits to the same place, tests still failing. The remedy is to force it to re-examine its diagnosis.

The remedies are *opposite*: applying the wrong one wastes the budget the monitor was meant to save. A monitor that cannot separate the modes cannot be acted on, however high its AUC.

We measured the two modes on 236,137 mechanically labelled edits across 25,681 runs on 1,213 real software-engineering instances, then replicated on two further disjoint sets of ~26,000 runs each:

failure mode	definition (mechanical, no judgement)	share
WRONG-FIX	failed, but it <i>did</i> edit a file the gold patch touches	67.0 / 69.0 / 66.4%
LOST	failed and never edited such a file	33.0 / 31.0 / 33.6%

Table 1: Two-thirds of failures happen with the agent already in the right file. Stable to within 2.6 points across three disjoint shard sets, and across model scale (66.8% Llama-70B, 70.7% 8B, 70.2% 405B).

The share barely moves between sets, so the split is a property of these corpora rather than of one shard set or one model size.

2 A measurement trap: self-referential localisation

We first reproduced the standard localisation measurement and obtained an apparent headline: **failed runs localise better than successful ones** (0.670 vs 0.594, $p = 3.3 \times 10^{-4}$). The result is an artefact of the target. That statistic is computed against **the run’s own final patch**, so both the numerator’s file set and the denominator come from the run itself: it measures convergence on what the run produced, not aim at the *correct* file, and a run that fixates on the wrong file and never wavers scores a perfect 1.0.

Recomputed against an independent gold patch for 927 of the 1,213 instances the direction reverses, and does so again on both held-out sets (0.492 vs 0.421, $p = 3.6 \times 10^{-3}$, $n = 198$; 0.512 vs 0.415, $p = 2.8 \times 10^{-5}$, $n = 229$ — paired within-instance tests). The control that settles it: restricted to runs whose own patch is a single file that *is* a gold file — where the two target sets must coincide — the metrics agree (0.615 vs 0.638, mean absolute gap 0.033 over 8,801 runs). Figure 1 plots both readings side by side.

	frozen (0–3)	held-out A (4–7)	held-out B (8–11)
on-target, own patch (<i>the trap</i>)	0.594 / 0.670	0.593 / 0.677	0.614 / 0.686
on-target, gold patch	0.477 / 0.407	0.492 / 0.421	0.512 / 0.415
ever reached a gold file	0.982 / 0.670	0.964 / 0.689	0.976 / 0.664

Table 2: Solved / failed, on three disjoint sets of runs (each ~26,700 runs, 1,213 shared instances). The trap inverts in all three; the independent target reverses it in all three.

Every quantity in this paper is built on the independent target. We tested three candidate explanations for the inversion (confident wrongness, patch breadth, fixation) and **all three failed**; the mechanism is recorded as open rather than asserted.

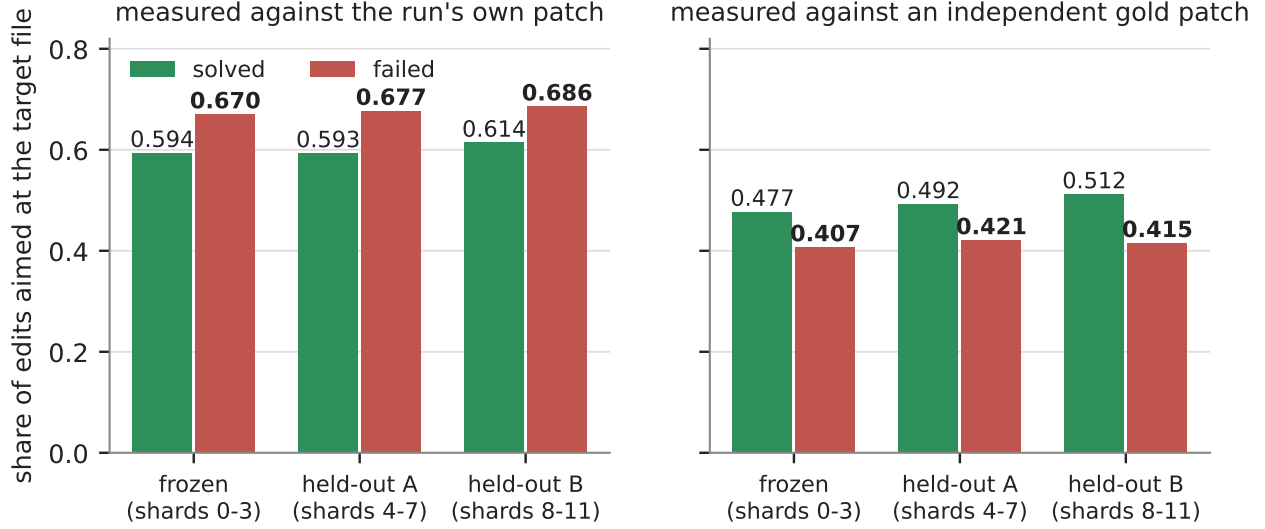


Figure 1: Left: measured against each run’s own final patch, failed runs look *better* at localisation than solved ones. Right: measured against an independent gold patch the ordering reverses, in all three disjoint shard sets.

3 The router

3.1 Overview of the approach

The router is a two-head classifier over a fixed-length feature vector computed from the first f fraction of a run, $f \in \{10, 20, 40, 60\}\%$. Both heads read the same inputs and answer different questions: $P(\text{fail})$ asks whether the run will fail, and $P(\text{wrong-fix} \mid \text{fail})$ asks whether a failing run has found the right file and is failing to fix it (WRONG-FIX) rather than still searching for it (LOST). The outputs combine into a single decision rule: intervene only when $P(\text{fail})$ is high, then apply the remedy the mode head implies — SEARCH for LOST, VERIFY for WRONG-FIX.

Three design choices carry the claim. Every feature is a function of the prefix alone, so the model is evaluated at the moment the decision would be made; run length is excluded because, at a fixed prefix fraction, it is not observable online. The mode question is asked on the same rows, folds and features as the failure question, so the comparison against published detectors isolates the question asked rather than the data or the model class. And the model is trained on one shard set and scored on shard sets it never saw, which makes the reported margin a property of the mechanism rather than of the fitting.

3.2 Runtime interface and outputs

`python demo/route.py --replay <run_id>` reads a run’s first f fraction of steps ($f = 10, 20, 40, 60\%$) and prints, at each fraction, $P(\text{fail})$, $P(\text{wrong-fix} \mid \text{failing})$, and the resulting intervention. On a real held-out run:

```
at 20% of the run:
P(fail) = 0.772   P(wrong-fix | this is a failing run) = 0.682
-> VERIFY (it is in the right place, make it check its fix)
ground truth: reward=0 -> WRONG-FIX
```

3.3 Input features and causal constraints

The design constraint that makes the result meaningful is **causality**. The prefix is steps $0 \dots L - 1$ with $L = \text{round}(f \cdot n_{\text{steps}})$; the run’s own length appears in *no* feature. A self-test overwrites every post-cutoff step and asserts all 112 feature columns are unchanged — it caught a genuine leak during development, now fixed. Labels come from the dataset’s gold patch, never from the agent’s output. Folds are task-disjoint.

The inputs are ordinary observables: edit counts and rates, no-op edit fraction, repeat and entropy of command signatures, the read/edit/run/verify mix, observation-size trend, and the pass/fail/error signals visible so far.

3.4 Controlling for run length

An earlier version compared the router against “position” and found it marginally ahead. That comparison was **invalid**: at a fixed prefix fraction, position *is* the run’s length — a quantity no online system can observe (verified: its AUC matches run length’s to ≤ 0.006). Against a length-matched control the learned model holds at 0.63–0.76 and strictly dominates run length; adding position to the features changes the result by ≤ 0.002 .

4 Results

4.1 Published detectors cannot identify the failure mode

Every published family is re-implemented on the identical rows and task-disjoint folds, so the comparison is across models on one matrix rather than across papers. On failure prediction the families are competitive; on the mode question they sit at or below chance. Figure 2 shows the same comparison cell by cell.

target	Route	position	AgentStop-style	loop / redundancy
will this run fail?	0.7145	0.6718	0.5589	0.45–0.58
LOST or WRONG-FIX?	0.7324	0.5993	0.5960	0.41–0.54

Table 3: Identical rows, identical task-disjoint folds. Averaged over 24 cells (every ordered pair of the three shard sets $\times$ four prefix fractions). On the mode question the published families carry *no information*.

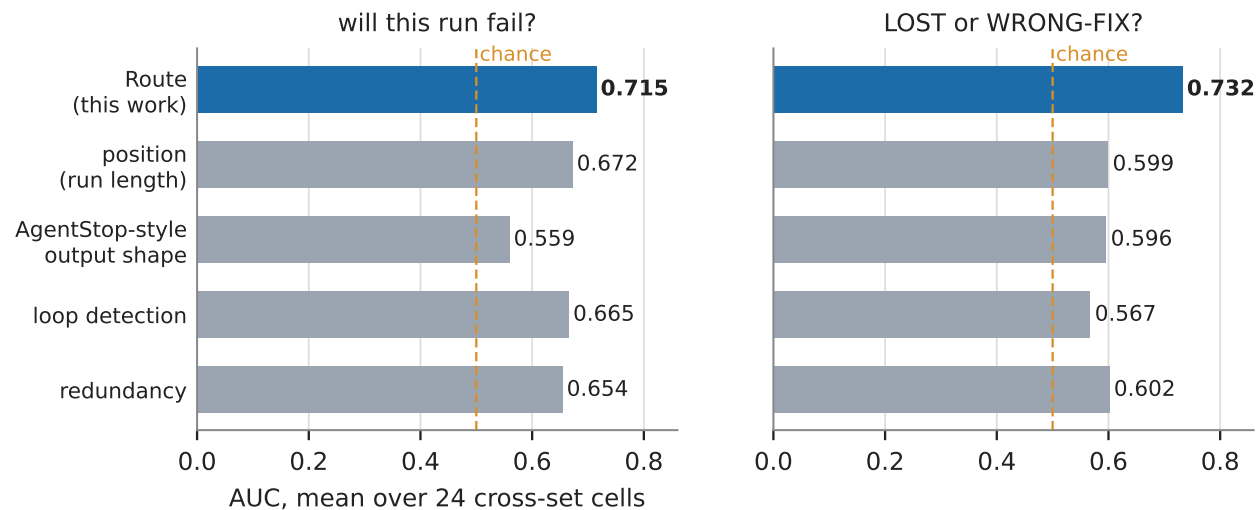


Figure 2: Means over the 24 cross-set cells. On the mode question every published family sits at or below the chance line while the router reaches 0.732.

4.2 Transfer to unseen shard sets

Direction by direction, at the deployed 20% checkpoint, the failure head scores 0.6965 (0–3→4–7), 0.7220 (0–3→8–11), 0.7086 (4–7→0–3) and 0.7079 (8–11→4–7); the mode head scores 0.7346, 0.7274, 0.7086 and 0.7317 on the same rows. Nothing here is evaluated on data the model was fitted on; Figure 3 shows the six ordered pairs against the within-set mean.

	will this run fail?	LOST or WRONG-FIX?
within-set (same shard set, task-disjoint folds)	0.7169	0.7117
cross-set (trained on one set, tested on another)	0.7145	0.7324
worst of the 24 cross cells	0.6752	0.6817
position baseline	0.6718	0.5993
AgentStop-style baseline	0.5589	0.5960
gain over the field	+0.1556	+0.1364

Table 4: Means over the 24 cross-set cells (three shard sets, every ordered pair, four prefix fractions) and over the three within-set cells. Cross-set performance *equals* within-set performance: the margin is not fitted to one corpus. The worst single cell still beats the best baseline’s mean.

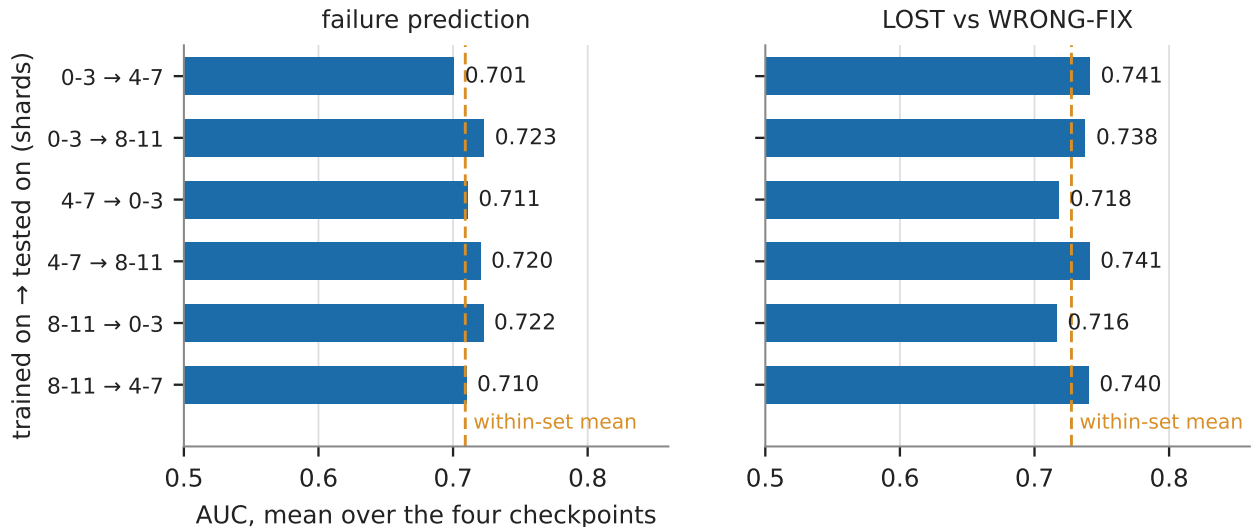


Figure 3: Each bar is a model trained on one shard set and scored on another; the dashed line is the within-set mean. Cross-set performance equals within-set performance.

4.3 Limits of cross-scaffold generalisation

The transfer reported in Table 4 is over *shards* of one scaffold: the same prompt, tool set and observation format. To test whether it survives a different scaffold we applied the same code path to three public corpora recorded from **other scaffolds** — OpenHands on SWE-rebench, a four-framework corpus, and SWE-Gym’s OpenHands trajectories — building the per-step frame with the same feature code for training and for test data, and adding a control that a near-chance result cannot supply by itself: how well the *same* features do when the model is fitted inside the target scaffold.

This is a negative result and we report it as one: **the transfer established here is shard-level generality within a scaffold, not scaffold-independent generality.** It is consistent with the structural finding in the limitations: a measurement that depends on one scaffold’s output format cannot be expected to survive a different one. The mode head could not be tested this way at all: the gold patches that define its labels exist for 227 of the 9,921 instances in these corpora that have edits, so the LOST/WRONG-FIX question has almost no negatives to score. Figure 4 sets the two columns against each other.

4.4 Calibration of the false-alarm rate

Our own earlier “zero false alarms at every budget” claim was **withdrawn as circular**: the detector and the label were the same statistic, so the reported numbers were those of an oracle. This version uses an explicit null (for the mode target, runs that are LOST; for failure, runs that succeed) and a sequential rule valid under arbitrary dependence:

test corpus (scaffold)	runs	fitted inside it	transferred from SWE-agent	best fixed baseline
SWE-rebench / OpenHands	67,074	0.653	0.495	0.655
thoughtworks agentic-coding	15,000	0.759	0.433	0.640
SWE-Gym / OpenHands	6,055	0.761	0.319	0.425

Table 5: AUC on identical rows; “fitted inside it” is task-disjoint 5-fold OOF. The features carry real signal inside every scaffold, and a model trained on SWE-agent carries almost none of it across. Three variants that drop the command-digest and observation-scale features (22–31 features) move the transferred column only within 0.32–0.54, so this is not a bridge artefact.

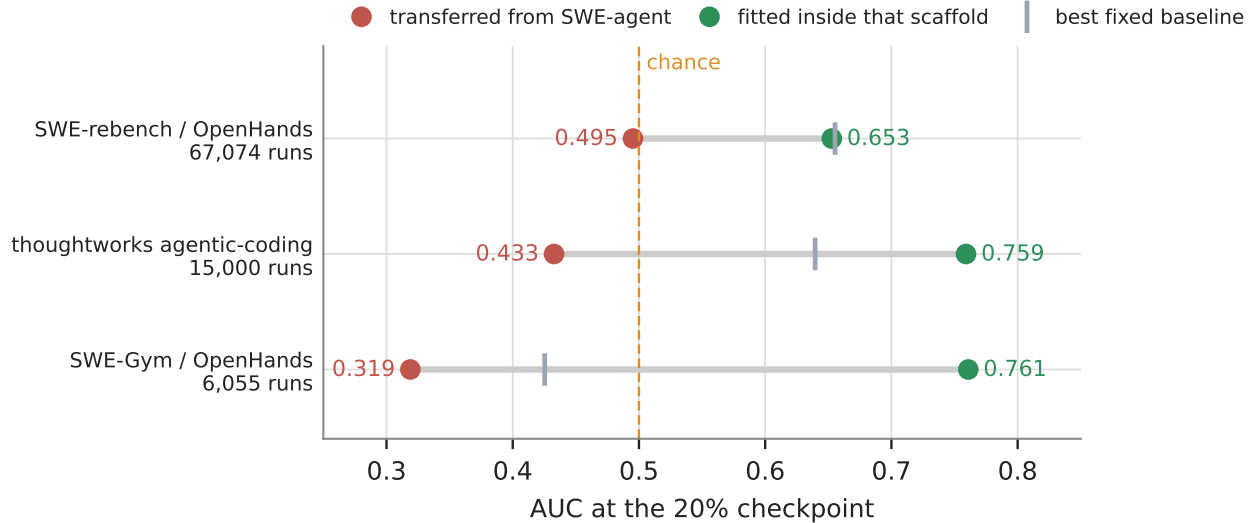


Figure 4: The same features fitted inside each scaffold (green) reach 0.65–0.76; transferred from SWE-agent (red) they reach 0.32–0.50, at or below the fixed baselines. The generality claimed in this report is therefore bounded to one scaffold’s shards.

8 of 8 levels controlled on the mean; Figure 5 plots the same comparison. A raw 0.5 threshold flags 98.4% of never-failing runs; the calibrated rule flags **3.39%**. We also report a negative: a formal sequential e-value test is valid and **never fires** at $\alpha \leq 0.05$.

4.5 Decision value against fixed policies

A detector is useful only if the decision it drives beats fixed policies that require no detector. The router is therefore scored against those policies on the runs that actually failed, giving 1 for a route that names the true failure mode and λ for one that does not, and sweeping both the checkpoint and the cost λ . It wins in **12 of 12** cells. The ordering of the fixed policies is itself informative: **always SEARCH scores below random** (0.518 against 0.656), the practical consequence of the 67% WRONG-FIX share — on this corpus a location is more often the wrong action than the right one. Figure 6 traces the whole sweep.

5 Live evaluation: verifier leakage and intervention effects

We ran the premise against live agents (DeepSeek V4.1 Flash, temperature 0, real Python packages, their own test suites as verifier, no Docker): **241 episodes** in five conditions over the same 48 tasks, **\$2.26 in total**.

In the first two conditions **every run reached the correct file** (100%), which would place the LOST mode outside this setting. The explanation is mechanical: pytest prints the failing test’s *filename*, and that filename contains the module name (`test_ioutils.py` → `ioutils.py`), so the defect’s location is readable

level	achieved false-alarm rate	detection
$\alpha = 0.05$	0.046 (worst split 0.061)	0.160
$\alpha = 0.10$	0.078	0.258
$\alpha = 0.20$	0.123	0.386

Table 6: Requested false-alarm rate against the rate actually achieved, and the detection that buys, across the calibrated levels.

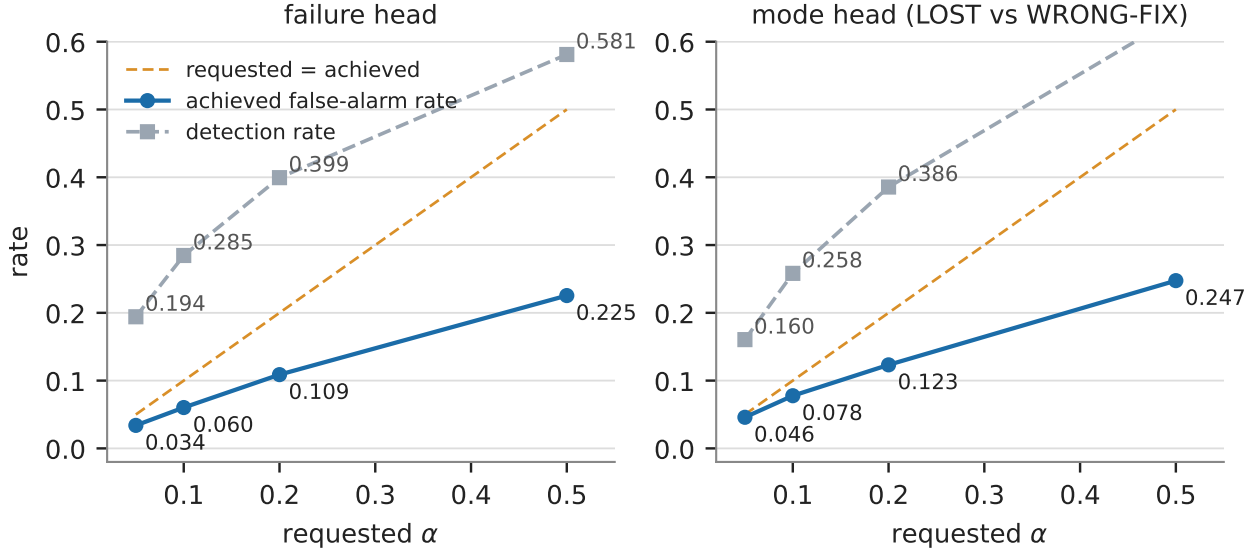


Figure 5: The sequential rule is controlled at every level on the mean; the detection rate it buys is modest and is reported as such.

off the failure message. **51.5% of test observations handed to the agent name the gold module explicitly.** Localisation in these benchmarks is string-matching, not diagnosis.

We therefore added a **masked** condition: the agent is told *that* tests fail (it still sees “3 failed, 516 passed”) but never *which* or *where*. The grader always sees the true output, so the success criterion is unchanged. Two further conditions test the interventions the router would actually prescribe: **hinted** hands over the file and function (the SEARCH action), and **verify** appends, after the second failing test run, the instruction to stop and re-check the diagnosis rather than edit again (the WRONG-FIX action). **Nudge** is the control for the second one: same trigger, same channel, same length, no content.

Two conclusions remain, both narrower than intended. The one solid causal result concerns the *benchmark*: hiding which tests failed costs 27.5 points, while the other conditions are statistically indistinguishable from each other. The runtime actions the router prescribes both point the right way — a hint or a forced re-check is worth roughly +0.2 over doing nothing, and the re-check costs no turn budget (11.1 turns against 12.2) — but **this experiment cannot attribute the gain to the instruction rather than to being interrupted**, because a content-free message at the same moment moves success almost as far. Separating the two needs a larger n than this suite can supply; we report the ambiguity rather than the attractive reading. Figure 7 shows the five conditions with Wilson intervals.

The router on live runs: a further negative. The frozen model was then applied unchanged to these episodes — a different scaffold, a different model, real packages — and it does **not** transfer: AUC **0.426** at the 20% checkpoint ($n = 125$) against 0.584 for the published output-overlap baseline. Re-fitting on the training corpus with only the features that survive the transcript bridge, and dropping the observation-scale family, recovers 0.602. This is the conclusion the three external corpora reached in §4.3, from a different direction, and it is why the claim in this paper is bounded to one scaffold’s shards.

Note: the masked condition’s first run produced 0/48 success and 0% reaching the gold file. That was a

policy (on runs that failed)	value ($\lambda = 0.25$)
Route (router-guided)	0.764–0.800
always VERIFY	0.732
random	0.656
always SEARCH	0.518

Table 7: Beats all three at every fraction and every cost setting — 12 of 12 cells.

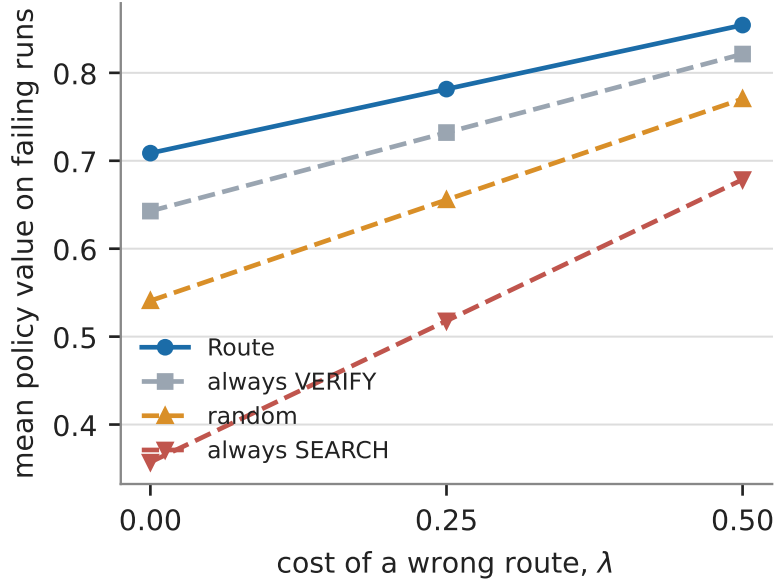


Figure 6: Scoring 1 for a correct route and λ for a wrong one, on runs that failed. The router beats always-verify, random and always-search at every checkpoint and every cost setting.

***harness fault, not a result:** every episode died before its first tool call because the recorded Python interpreter had been deleted from a temp directory between runs, and no tokens were spent. It is recorded because the number has the shape of a strong finding and was disbelieved only by reading the transcripts. The environment was rebuilt in-repo and the condition re-run; the numbers above are the corrected ones.*

Harness faults that looked like results. (i) **65.2% of turns were unparseable** in the first pilot — our protocol was ad-hoc text, but the model emits its own XML tool-call format regardless; native function-calling took this to 0–4%. (ii) **The agent was writing its own tests into `tests/`**, which pytest collects, so an agent-authored passing test could have scored a run as successful *without the defect being fixed*; the verifier is now restored to the packages’ originals before grading, and in 72 of 84 valid episodes the agent had modified test files (1,076 files removed). (iii) the deleted interpreter, above.

6 Reproducible demonstration

```
python demo/route.py --fit # train on shards 0--3, cache the model
python demo/route.py --summary # router vs baselines on held-out shards
python demo/route.py --list # held-out runs with their true modes
python demo/route.py --replay <run_id> # step through one run, turn by turn
```

Offline, deterministic, no API key; reproduces the held-out numbers on demand.

condition	what the agent sees	success	vs unmasked	reached the file
masked	only “N failed, M passed”	0.286 ($n = 42$)	-0.086 ($p = 0.490$)	0.952
unmasked	full pytest output (normal CI)	0.372 ($n = 43$)	—	1.000
nudge	full output, plus a content-free “keep going” after the 2nd failure	0.452 ($n = 42$)	$+0.080$ ($p = 0.512$)	1.000
hinted	the exact file and function	0.561 ($n = 41$)	$+0.189$ ($p = 0.125$)	1.000
verify	full output, plus a re-check-your-diagnosis instruction, same trigger	0.571 ($n = 42$)	$+0.199$ ($p = 0.084$)	1.000

Table 8: Five conditions, same 48 tasks, valid episodes only (241 raw, 31 dropped because the mutated package did not fail its suite at episode start, which makes success meaningless). The **only** differences that reach significance are the ones involving *masked*: against hinted ($p = 0.015$) and against verify ($p = 0.015$). Both runtime interventions land at 0.56–0.57 against 0.372 for no intervention, but neither is individually significant at this sample size, and the content-free control at the same trigger (0.452) is **not** distinguishable from the instruction it controls for ($p = 0.383$).

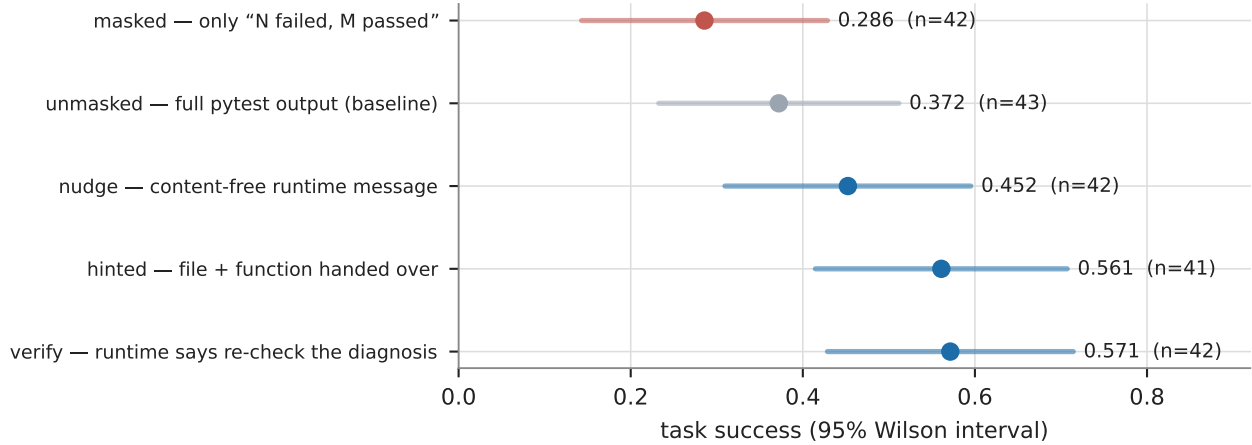


Figure 7: Only the comparisons involving *masked* reach significance. Both runtime interventions point the right way but neither is individually significant at this sample size.

7 Limitations

- **The cross-scaffold limitation is severe, and measured rather than assumed (§4.3).** Every mechanical measurement rests on a structural footer SWE-agent prints into its observations (`[File: ... (N lines total)]`) in 95.8% of edit steps. That footer is **absent from every other scaffold tested** — OpenHands, the PI agent, mini-swe-agent-plus and a multi-framework corpus all show 0% — and the failure head trained on SWE-agent scores **0.319–0.495** on 88,000 runs from three other scaffolds. **Scaffold-independent generality is not claimed and this is the evidence against it.**
- **The dead-end rate is scaffold-specific, not a property of coding agents.** Four other scaffolds, measured by the same mechanical rule: OpenHands **3.7%**, the PI agent **11.8%**, a multi-framework corpus **17.9%**, mini-swe-agent-plus **23.6%** — a factor of six, and it moves in both directions away from SWE-agent. **19.1% is SWE-agent’s number and is not claimed for coding agents in general.**
- **The 3.7% figure rests on patch recovery from the transcript** for a scaffold with no printed patch; reported as a discrepancy to investigate.
- **Step-level telemetry does not track what a person calls progress.** An independent blinded reader judged 12 windows (pre-registered, sealed key, one pass): agreement with the stored reader-produced judgements was **83.3%** (Wilson [55.2, 95.3], $\kappa = +0.667$) but with the mechanical progress measure only **50.0%** ([25.4, 74.6], $\kappa = +0.100$), with the disagreement concentrated on windows where the workspace moved without advancing the task. The labels used in *this* paper do not depend on that measure — they

are defined against the dataset’s gold patch — but the pilot is why we do not claim that telemetry tracks progress, and it bounds what any behavioural monitor of this family can be expected to detect.

- **The mode labels are coarse.** “Ever touched a gold file” is a proxy, not a judgement of fix quality; gold matching is basename-based and 4.10% of edit steps carry no filename, which biases LOST upward.
- **The live experiment is small** ($n = 41\text{--}43$ per condition, five conditions). The only comparisons that reach significance involve the masked condition (withholding the failure location costs 0.275, $p = 0.015$). Both runtime interventions point the right way (+0.189 for the hint, +0.199 for the forced re-check) but neither is individually significant ($p = 0.125$, $p = 0.084$), and the content-free control is not distinguishable from the instruction it controls for ($p = 0.383$). The experiment is reported as *suggestive of the interventions and conclusive only about the benchmark*.
- **The inversion’s mechanism is unexplained:** three candidates tested, all failed.
- **Temperature-0 inference is not deterministic** ($\sim 9\%$ of per-instance outcomes flip across repeats), so within-instance comparisons are reported with runs-per-instance (mean 21.9).
- **The corpus is 80,035 runs in three disjoint tables**, not one merged table. The single-table rebuild was attempted, failed, and is recorded.

8 Automated verification

Six automated gates run on every change: invariant tests 11/11; cross-artifact consistency passes; 28/28 headline claims trace to their artifacts; 0 mismatches against the one-page summary; every cited artifact exists; and 65/65 paper numbers trace to the artifact each one names. The last of these was written during this work and **found five real errors in our own manuscript**: values quoted as pooled that were within-instance, figures mixed across two artifacts’ row sets, and a transfer table whose cells had been read off the wrong rows. All are fixed. Complete data package: `research/EXPORT/`.

9 Use of AI tools

A language model (`deepseek-v4.1-flash`) was used as a research assistant throughout: writing analysis code, running statistical tests, retrieving and verifying literature, and drafting text. Every load-bearing citation was verified by hand against the primary source, and two reconnaissance claims were rejected as unverifiable. The model also served as the *subject* of the live experiment. It was not used to generate data; all measurements come from public corpora and real test suites executed locally. **Eleven of its own claims were falsified and retracted**, documented pass by pass in the project’s assistance log. Full disclosure, including the required chat records, is in the separate acknowledgement and disclosure document.

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致谢与人工智能使用声明

Acknowledgement and Disclosure of AI Use

一、研究背景

本项目研究编码智能体（coding agent）的失败方式，以及运行时系统能否在失败发生之前判断智能体究竟是「找不到代码」（LOST）还是「找到了代码但修改不正确」（WRONG-FIX）——因为这两种情况需要相反的干预措施。研究使用的数据全部来自公开的智能体运行轨迹数据集（SWE-agent / SWE-rebench、Terminal-Bench 2.0，以及另外四个开源脚手架），以及八个开源 Python 软件包在本机运行的测试套件。不涉及人类受试者，不使用任何非公开数据。

二、指导老师与参赛学生的关系

[待填写] 四项缺一不可：

- 老师姓名与所在单位：[待填写]
- 学生与老师的结识渠道：[待填写]
- 指导有偿还是无偿（有偿须写明费用与来源）：[待填写]
- 老师参与与未参与的部分：[待填写]

三、分工说明

本节写清每位作者的具体贡献。下述两项工作属于作者二（Xuhao Chen）而非外部协助，因此在此列出，而非列入「来自他人的帮助」。

- 选题来源：[待填写]
- 数据获取：[待填写]
- 分析与计算：[待填写]
- 实验设计与实施：[待填写]
- 独立的盲评人机对照试验（作者二 Xuhao Chen）：由作者二设计并实施——编写盲评窗口包与评分脚本（仅用标准库，含 Wilson 区间、 κ 、McNemar 检验）、预先注册判定规则、密封答案并在全部判定提交后才开封。结果为：人工判定与读者标注一致率 83.3% ($\kappa = +0.667$)，与机械指标一致率仅 50.0% ($\kappa = +0.100$)。
- 数据重复问题的发现与修正（作者二 Xuhao Chen）：由作者二发现 Terminal-Bench 2.0 数据集中存在重复试验条目并给出去重规则；据此重建数据表，并重新计算受影响的全部统计量。
- 撰写论文：[待填写]
- 遇到的困难及解决经过：[待填写]

四、人工智能使用情况

工具与版本：deepseek-v4.1-flash（大语言模型），通过 DeepSeek Harness (DSH) 自主编码助手接口调用。每一次模型调用的时间戳、token 数与费用均记录在 `research/results/rebuild/llm_ledger.jsonl` 中；活体实验（241 次运行、5 种条件、48 个任务）共花费 **2.26 美元**。

使用阶段与用途：（1）文献检索与核实（每条关键引文均由人工回原文献核对，其中两条前期结论因无法查证而被剔除）；（2）编写轨迹解析与数据表构建代码；（3）编写并运行统计分析（任务不交叉的交叉验证、自助法区间、Wilcoxon 检验、校准曲线、决策曲线）；（4）在活体实验中，模型本身是被测试的对象；（5）编写初稿文字，由参赛学生审阅并重写。

时间与频率：2026 年 9 月 10 日至 9 月 13 日，共四个工作时段，累计 34 次，逐次记录于项目内的 AI 使用日志（`AI_ASSISTANCE_LOG.md`）。

聊天记录：按组委会要求另行提交。

五、来自他人的帮助

[待填写] 本节只填两位作者之外的帮助；确无此类帮助，请写明「无」。（盲评试验中被判定的 12 个窗口，其判定人若在两位作者之外，请在此写明姓名。）

六、关于 AI 出错的自愿声明

为如实说明研究的可信度，我们列出 AI 助手自身被推翻的十一项结论，包括：一个自指的定位指标使结论方向相反；一项「零误报」声称属于循环论证；一项基线实际上等于运行长度；一次活体实验出现「48 次全部失败」的惊人结果，实为解释器被删除的实验故障；论文中一张移植结果表引用了错误的数字行（由自动审计脚本发现）；以及 Terminal-Bench 数据集的重复条目导致运行数与步数被高估（34,029 与 1,073,923 修正为 29,103 与 887,137，由合作者发现）。这说明本项目的规则是：只有当试图推翻某结论的检验失败时，该结论才被保留。全部十一项均逐次记录在 `research/AI_ASSISTANCE_LOG.md` 中。

七、声明

我们声明：本文所述研究成果为参赛学生的原创工作；引用他人成果均已注明并在参考文献中列出；上述人工智能使用情况完整、真实；我们已知悉并遵守本届竞赛的学术诚信规定。

参赛学生签名： _____

Ziheng Yu

Xuhao Chen

指导老师签名： _____

日期： _____